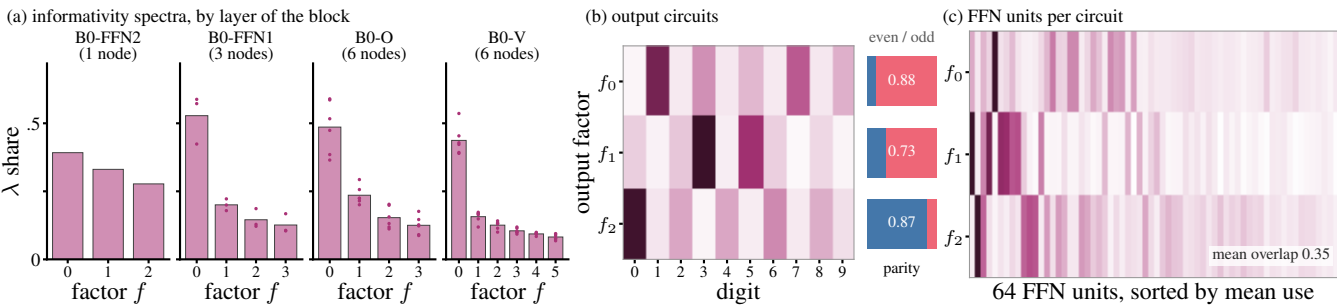
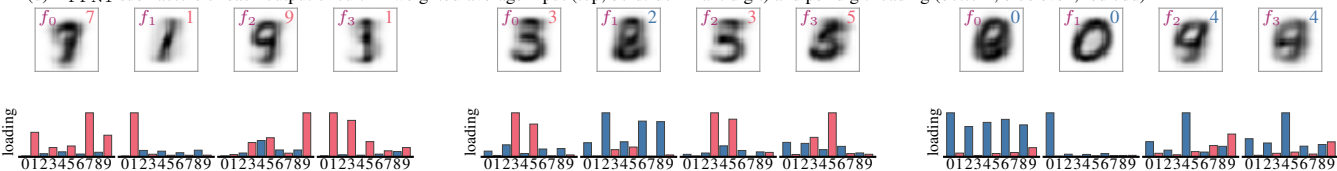
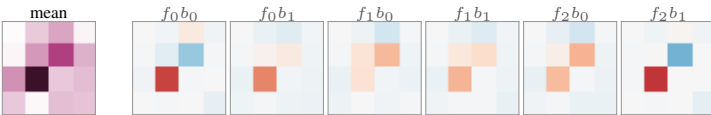




(d)  $L_{\text{FFN1}}$  sub-factors of each output circuit — weighted-average input (top, bold: dominant digit) and per-digit loading (bottom; blue even, red odd)



(e) CLS attention per factor



CLS-token weight 0.027; deviation of each leaf  $k_0$  from the mean  
(all 36 factor maps: pairwise  $\cos \geq 0.97$ )

(f) strongest stimuli per circuit

